

Pranav P

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Portfolio: [aikkaraportfolio.vercel.app](#) · **LinkedIn:** [linkedin.com/in/pranavaikkara](#)

PROFESSIONAL SUMMARY

Data Scientist and GenAI Engineer building production LLM systems end-to-end — agentic RAG pipelines, reliable voice agents, ingestion infrastructure, and fine-tuned domain models deployed on AWS. I pick the right retrieval approach for the job (vectorless tree-walk when documents have real structure and accuracy matters more than speed; semantic similarity when it doesn't) and ship small fine-tuned models that outperform frontier LLMs on narrow tasks. Strong classical ML background — time-series, satellite imagery, and explainable models for high-stakes decisions — which informs how I evaluate AI systems beyond surface benchmarks.

WORK EXPERIENCE

FarmwiseAI Pvt. Ltd. — Perungudi, Chennai

Associate Data Scientist

April 2025 – Present

- **Vectorless-RAG agentic system:** Built a PageIndex-style retrieval pipeline for internal documents — an LLM walks a tree-of-contents and answers from selected nodes, without embeddings or a vector database. Migrated off the original Qdrant + bge-large pipeline after cosine similarity kept returning text that *looked like* the question rather than text that *answered* it; the tree-walk approach delivered noticeably better answers on long structured documents (policies, runbooks, product specs).
- **Production voice agents:** Built reliable STT ↔ LLM ↔ TTS loops with per-hop latency budgets, interrupt handling, and a state machine that handles mid-response interruptions cleanly. Solved two production issues with a companion-agent pattern: an *auto-compact agent* that summarizes context every 200k–300k tokens (preventing drift from conversation bloat), and a *parallel tool-calling agent* that runs tools while the voice agent is still speaking (so the user never hears a pause on lookups).
- **Universal ingestion pipeline:** Built a pipeline that normalizes PDFs, scans, audio, spreadsheets, and images into clean, LLM-friendly structured context. Each input type is routed to the right normalizer (OCR for scans, STT for audio, table parsers for spreadsheets) so downstream retrieval doesn't need to special-case format. OCR runs LightOnOCR-2 1B on AWS SageMaker, served through vLLM with PagedAttention and KV caching for high throughput.
- **Internal intelligence platform:** Stood up a company-wide assistant on OpenWebUI, RAG-backed against curated internal sources (policies, docs, onboarding material). Self-hosted, with a swappable model backend (Groq, local, etc.) that doesn't require changing the UI.
- **Fine-tuned geospatial small models on AWS:** Deployed specialized small models on SageMaker for location, terrain, and coordinate queries where frontier LLMs hallucinate. Called as a tool from the main agent when the query is geospatial — much better accuracy and lower cost than routing every geo query through a GPT-4-class model.
- **Agentic chatbot with Google ADK + MCP:** Built a chatbot using Google ADK, integrating MCP tools (database queries, Google Search, Qdrant-backed RAG). Sub-agent routing to smaller models for lightweight tasks, Pydantic validation on every LLM response, and WebSocket streaming for live execution updates.
- **Satellite imagery time-series:** Processed one year of Tamil Nadu agricultural field imagery stored on Azure — pixel-level vegetation signals extracted across time using distributed PySpark jobs so a year of scenes could be processed in parallel rather than field-by-field, then Dynamic Time Warping (DTW) to align growth curves across fields planted at different dates. Used to infer sowing dates for fields without explicit metadata.

- **Tamil LLM tokenization:** Trained a BPE tokenizer on a Tamil text corpus to extend an open-source LLM's Tamil understanding. Used a custom tokenizer because English-trained general tokenizers fragment Tamil text into far more tokens than necessary, reducing efficiency and effective context length.
- **Cross-cutting stack:** **LiteLLM** as the LLM gateway across every project (provider swaps without rewriting calling code). **Langfuse** for end-to-end observability — every LLM call, trace, latency, and cost logged so production debugging becomes a diff, not guesswork. **MCP** as the standard protocol for connecting tools between agents.

Infosys — India, Remote

Data Visualization and Analyst Engineer Intern

February 2025 – April 2025

- Worked on data visualization and analytics to transform complex datasets into actionable insights for non-analyst stakeholders.
- Built interactive dashboards and visual reports using Power BI and Python.
- Performed data preprocessing, exploratory data analysis (EDA), and trend analysis to surface key patterns from real-world datasets.

Descpro Pvt. Ltd. — Palakkad, Kerala

Machine Learning Developer Intern (NBFC Loan Approval System)

December 2023 – April 2024

- Developed an AI-powered system to automate loan approval and monitor customer repayment patterns for a non-banking financial company (NBFC).
- Built and deployed machine learning models to evaluate loan eligibility based on income, age, expenses, and other risk parameters.
- Designed an individual customer monitoring model using LSTM to predict repayment delays at the per-customer level.
- Built a loan inquiry forecasting model to predict upcoming application volumes, supporting operations capacity planning.

Thapovan Info Systems Pvt. Ltd. — T. Nagar, Chennai

Machine Learning Intern (Predictive Disease Modeling)

September 2023 – November 2023

- Built a heart attack risk model from raw clinical data — translating a clear medical need into a defensible predictive system.
- Curated and engineered 20+ key health indicators into a model-ready dataset, improving data quality before training.
- Applied SHAP and LIME (explainable AI) so doctors could see *why* the model flagged any individual case, not just the score — important in a setting where the prediction has to be defensible to a clinician.
- Built an LSTM-based alert system to flag time-series patterns suggesting rising risk, giving doctors earlier warning windows.

TECHNICAL SKILLS

- **GenAI & LLM Systems:** RAG (vectorless + traditional), Agentic Systems, LangChain, LangGraph, Google ADK, Model Context Protocol (MCP), Tool Calling, Pydantic AI / Instructor, PageIndex, Qdrant, FAISS, HuggingFace, vLLM, LiteLLM, OpenRouter, Ollama, Langfuse (observability), Unsloth, QLoRA
- **Machine Learning & Data Science:** Random Forest, LSTM, Time-Series Analysis, Dynamic Time Warping (DTW), Explainable AI (SHAP, LIME), Data Interpolation, Data Leakage Prevention, Feature Engineering
- **Backend & AI Infrastructure:** Python, FastAPI, Pydantic, WebSockets, Streamlit, AWS SageMaker, Azure, llama.cpp, Vercel, Git
- **Data & Analytics:** SQL, Power BI, Pandas, NumPy, PySpark, Satellite Imagery Processing

PROJECTS

Domain-Specific Loan Officer Agent · Unsloth, QLoRA, Gemma-2B, llama.cpp

- **Fine-tuning:** Fine-tuned Gemma-2B on a synthetic instruction dataset using QLoRA and Unsloth for memory efficiency, training a compliance-aware tone for the domain.
- **Edge deployment:** Quantized to GGUF and deployed via llama.cpp for low-latency CPU inference — proof that a well-fine-tuned small model is a real production path for narrow domains, no GPU required.

Companion AI — Voice-Enabled ICU Support System · PyTorch, Transformers, STT/TTS

- **Pipeline:** Integrated real-time Speech-to-Text with BERT-based sentiment analysis to detect emotional distress in ICU patients, triggering context-aware empathetic responses via TTS. Built for patients who are alone, disoriented, and unable to use a phone or tablet — voice is the right modality for them.

Agentic Recruitment Platform · Google Opal, Selenium, LangChain

- **Automation:** Built an autonomous agent that scrapes job boards and uses Google's experimental Opal model to rewrite resumes in real-time, mapping candidate skills to job descriptions via a RAG pipeline. Most of the engineering went into error recovery — stale selectors, captcha walls, rate limits — not the LLM call itself.

Portfolio Site with Vectorless-RAG Chatbot · Vercel, Groq, PageIndex, Node.js

- **The chatbot is the demo:** The portfolio site (linked at the top of this résumé) is itself a vectorless-RAG application. An LLM walks a tree-of-contents of a hand-authored markdown knowledge base about my work and answers from selected nodes — the reasoning panel under each answer shows which nodes were chosen, so the retrieval approach is a live demo rather than just a claim on the page.
- **Zero-cost infrastructure:** Single Node serverless function on Vercel, Groq Llama-3.3 for inference, no vector store, no database. Runs entirely on free tiers — zero dollars per month.

EDUCATION

Vellore Institute of Technology — Chennai, India	2021 – 2025
B.Tech in Computer Science with Specialization in AI and ML · CGPA: 7.98	
Palghat Lions School — Palakkad, Kerala	2020 – 2021
XII · 86.8%	

CERTIFICATIONS

• NPTEL Entrepreneurship	• Power BI Certificate
• TATA Data Visualization: Empowering Business with Effective Insights	• OCI Gen AI Professional Course
• Internship at Thapovan Pvt Ltd	• Infosys Springboard Internship — TechA Data Analytics using Power BI Foundation
	• Cisco Packet Tracer